

COEXTEND AI AUTOMATION ENGINEER — PILOT PROJECT

Prospect Intelligence MVP • Paid capability assessment

1. Pilot problem statement

Coextend wants an AI-assisted prospect intelligence system for its façade estimating, take-off, BOQ, tender support, shop drawing, BIM and technical support services. The founder currently spends significant time researching companies and deciding which prospects deserve attention.

- Input: a prospect/company name, website and optionally a contact.
- Research the company and relevant public information.
- Identify relevant decision-makers and current/recent projects or buying signals.
- Map evidence to likely Coextend service fit.
- Calculate the approved 100-point lead score.
- Produce a concise research brief with citations/source URLs and confidence labels.
- Prepare a CRM-ready JSON/CSV record and a recommended next action.
- Optionally draft a short email and LinkedIn message; LinkedIn actions must remain human-controlled.

2. Scope — build only this

Component	Required output	Must-have
Research engine	Structured company/contact/project findings + sources	Yes
Knowledge/RAG layer	Uses provided Coextend documents and returns source-grounded internal guidance	Yes
Lead scoring	100-point score + breakdown + priority band	Yes
Research brief	3–5 minute founder-ready output	Yes
CRM-ready output	JSON/CSV or API payload mapping to HubSpot-style fields	Yes
Outreach draft	Email + LinkedIn draft from research	Recommended
Simple UI/API	Minimal usable interface or FastAPI endpoint	Yes
Automated LinkedIn actions	Connection/message sending/scraping	No — prohibited for pilot
Production HubSpot deployment	Live production changes	No — mock/sandbox only unless separately approved

3. Timebox and working method

Target: 5 working days / 25–35 hours. If the candidate needs more time, they should state the reason and revised scope before continuing.

Day	Expected work	End-of-day evidence
Day 1	Understand brief; propose architecture; define data model, tools, risks and test plan.	Architecture note + repository skeleton.
Day 2	Build research ingestion/retrieval + public-source research flow.	Working research output on 3 test prospects.

Day	Expected work	End-of-day evidence
Day 3	Implement lead scoring + structured output + CRM-ready schema/API.	Scoring tests + JSON/CSV output.
Day 4	Add outreach drafts, basic UI/API, logging, error handling and evaluation.	End-to-end demo candidate build.
Day 5	Polish, test, document setup, limitations and next-phase roadmap.	Final demo + repo + README + test report.

4. Inputs Coextend will provide

- Company/services knowledge documents already created for the COEXTEND Growth Engine.
- Ideal Customer Profile and sales positioning.
- 100-point lead scoring rules.
- Research criteria and output template.
- One or more anonymised/sample CRM records if required.
- A small approved list of test prospects or permission for the candidate to use public company research for the pilot.

5. Expected technical approach

The candidate may choose the stack. We are evaluating engineering judgement, not forcing a particular framework.

- Use an LLM with structured outputs.
- Use retrieval/RAG for Coextend's internal knowledge rather than putting all knowledge into one prompt.
- Keep external research separate from internal company knowledge.
- Store source URLs and evidence snippets/metadata for important claims.
- Use deterministic scoring logic for the 100-point model instead of asking an LLM to 'guess' a score.
- Create a clear adapter/interface for future HubSpot integration.
- Include logging, retries, validation and failure handling.
- Keep secrets in environment variables/secret storage; never hard-code credentials.

6. Required research output

Section	Required
Prospect snapshot	Company, website, geography, business type, relevant services
Decision-makers	Name, title, why relevant, public source
Projects / signals	Relevant projects, tenders, hiring/growth/news signals, sources
Potential requirements	Likely Coextend service, evidence, confidence
Pain-point hypotheses	Evidence-based hypotheses, clearly labelled
Lead score	100-point breakdown + priority band
Recommended approach	Primary service, outreach angle, CTA
Sources	URL + source title + date where possible
Unknowns	What still needs human verification
Next action	Single recommended next action

7. Testing requirements

- Test at least 5 prospects, including at least 1 obviously poor-fit prospect.
- Include at least 2 cases where information is incomplete or ambiguous.
- Show at least 1 example where the system refuses to invent missing information.
- Show score repeatability: same inputs should produce the same deterministic score.
- Show at least one retrieval/citation test proving that internal Coextend knowledge is grounded in provided documents.
- Show at least one failure test (bad URL, timeout, missing field, malformed response) and how the system handles it.

8. Deliverables

- Working MVP (local or sandbox deployment is acceptable).
- Source code in a private repository.
- README with installation, configuration, architecture and run instructions.
- Architecture diagram.
- Environment/configuration template with secrets omitted.
- Sample input and output files.
- Test cases and results.
- Known limitations and technical debt.
- Estimated monthly running cost at 100, 500 and 2,000 prospects/month.
- 30/60/90-day recommended roadmap.

9. Candidate instruction

Do not aim to impress with a large number of features. Aim to demonstrate that you can build one small, reliable business workflow end-to-end. Use sensible engineering trade-offs, document assumptions, and call out anything that you believe should be changed before production.